

LEARNING PLAN

Unit Title: INSTALL TRUNKING SYSTEM	Unit code: 0713 251 24A
Name of Trainer: MARGARET WAITHERA	
Institution: ABERDARE TECHNICAL COLLEGE	Level: 5
Date of preparation: 05/05/2026	Date of revision: 05/10/2026
Number of learners: 3	Class: MODULE ONE
Skill or job task 1. Identify trunking accessories 2. Prepare trunking work pieces 3. Mount trunking work pieces 4. Install electrical cables and accessories 5. Perform Tests and Inspection 6. Promote Ethical Work Practices and Values	
Benchmark or Criteria to be used 1. Identify trunking accessories 1.1 Trunking types are identified as per IEC standards. 1.2 Trunking sizes are identified as per IEC standards. 1.3 Trunking accessories are identified as per IEC standards 2. Prepare trunking work pieces 2.1 Safety measures are applied as per OSHA and EHS standards 2.2 Electrical tools and equipment are selected as per work requirement 2.3 Measurements are taken as per work requirement. 2.4 Trunking work pieces are cut as per the measurements. 2.5 Housekeeping activities are performed as per the work requirement.	

3. Mount trunking work pieces
 - 3.1 Safety measures are applied as per OSHA and EHS standards.
 - 3.2 Mounting points are marked out as per IEC standards
 - 3.3 Trunking work pieces are mounted as per the IEC standards.
 - 3.4 Housekeeping activities are performed as per the work requirement.
4. Install electrical cables and accessories
 - 4.1 Safety measures are applied as per OSHA and EHS standards.
 - 4.2 Electrical cable colour code is identified as per IEC standards.
 - 4.3 Electrical cable sizes are identified as per IEC standards.
 - 4.4 Cables are laid as per the IEC standards.
 - 4.5 Accessories are fixed as per IEC standards.
 - 4.6 Housekeeping activities are performed as per the work requirement.
5. Perform Tests and Inspection
 - 5.1 Visual inspection is performed as per work requirements
 - 5.2 Continuity test is performed as per IEC standards
 - 5.3 Polarity test is performed as per IEC standards
6. Promote Ethical Work Practices and Values
 - 6.1 Personal management is demonstrated through self-awareness, self-esteem, emotional intelligence, stress management and assertiveness based on scope of work.
 - 6.2 Policies and guidelines are observed as per the workplace requirement.
 - 6.3 Self-worth and professionalism is exercised in line with personal goals and organizational policies
 - 6.4 Code of conduct is observed as per the workplace requirement.
 - 6.5 Teamwork is applied as per work place r Conflicts are resolved between team members in line with organization policy.

Week	Session No	Session Title	Specific Learning Outcomes	Learning Key Points	Trainee Activities	Resources & References	Learning Checks / Assessments	Reflections & Date
1	1&2	Introduction to Trunking	By the end of the session, the				Knowledge Checks:	

		Systems and Workplace Safety	trainee should be able to: 1. List the fundamental importance of workplace safety in electrical installations. 2. Identify basic workshop safety rules and their application. 3. Describe various types of Personal Protective Equipment (PPEs) and their uses.	- Significance of safety in electrical work - Potential hazards and risks - General conduct and practices - Emergency procedures - Types of PPEs for electrical work (e.g., gloves, safety glasses, hard hats) - Correct usage and maintenance of PPEs	- Engage in a Group Discussion to identify potential electrical hazards in a workshop setting and their importance. - Participate in a Q&A Sessions to clarify and internalize basic workshop rules. - Conduct a Gallery Walk to examine different types of PPEs and discuss their specific applications. <u>Follow up Activity:</u> 1. Research and list 5 additional safety precautions specific to	- OSHA and EHS safety guidelines - Workshop safety manual - Samples of various PPEs	-Oral questioning on workplace safety -Oral questioning on workshop rules. -Short quiz on PPE identification. Attitudes: -Demonstrate responsibility towards safety practices. - Show awareness of potential hazards.	
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					trunking installation not covered in class. (5 Marks) Due date:			
1	3&4	Trunking Types and Sizes	By the end of the session, the trainee should be able to: 1. Differentiate between various trunking types based on material and application. 2. Identify standard trunking sizes as per IEC standards.	- PVC trunking characteristics and uses - Metallic trunking characteristics and uses - Galvanized iron trunking characteristics and uses - Common dimensions (e.g., 16x16mm, 25x16mm, 25x25mm, 40x25mm) - Understanding IEC standards for sizing	- Conduct a Think-Pair-Share activity to distinguish characteristics and applications of PVC, Metallic, and Galvanized Iron trunking. - Engage in a Brainstorming session to list and categorize standard trunking sizes according to IEC standards.	- Samples of PVC, metallic, and galvanized iron trunking - IEC standards document (excerpts) - Installation manuals - Charts showing trunking dimensions	Knowledge Checks: -Practical identification of trunking types. -Written quiz on standard trunking sizes.	

			3. Select appropriate trunking types and sizes for given installation scenarios.	- Factors influencing choice of type (e.g., environment, load, aesthetics) - Factors influencing choice of size (e.g., cable capacity, space constraints)	- Participate in Case Studies where different installation scenarios are presented, requiring trainees to select optimal trunking types and sizes. <u>Follow up Activity:</u> 1. Sketch 3 different trunking installation scenarios and specify the ideal trunking type and size for each. (10 Marks) Due date:		-Practical identification Attitudes: -Exhibit attention to detail in material selection. -Value adherence to industry standards.	
	5&6	Trunking Accessories	By the end of the session, the trainee should be able to: 1. Identify	- Sockets and	- Engage in a	- Assorted trunking	Knowledge Checks: -Practical	

			various trunking accessories used in electrical installations. 2. Describe the function and application of each trunking accessory. 3. Select the correct accessories for different trunking system configurations.	switches compatible with trunking - Lamp holders and joint boxes for connections - Pattress boxes for surface mounting - Circuit breakers and Consumer Control Units (CCU) - Cut-outs, energy meters, earth rods, switch fuses - How accessories connect to trunking systems. - Considerations for placement and securing	Gallery Walk showcasing various trunking accessories, discussing their form and function. - Participate in a Jigsaw activity where groups become experts on different accessory categories, then share their knowledge. - Conduct a Concept Mapping exercise to illustrate the interconnections and placement of accessories within a complete trunking system. <u>Follow up Activity:</u>	accessories (sockets, switches, lamp holders, joint boxes, pattress, circuit breakers, CCU, cut-outs, energy meters, earth rods, switch fuse) - Product catalogs - Circuit diagrams	identification of accessories. -Oral questioning on accessory functions. - Practical demonstration Attitudes: -Demonstrate precision in component selection. -Show initiative in learning new components.	
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					1. Draw a simple trunking layout for a single room, clearly labeling all necessary accessories and their positions. (10 Marks) Due date:			
1	7&8	Electrical Tools and Equipment	By the end of the session, the trainee should be able to: 1. Identify various cutting, fastening, and measuring tools used in trunking installation. 2. Describe the correct application and safe handling of each tool.	- Hacksaws, trunking cutters, knives - Safe cutting techniques and material considerations - Screwdrivers, pliers, hammers, drills for fastening - Tape measures, try squares, spirit levels for accurate measurement - Multimeters for	- Participate in Demonstrations with Participation on the safe and effective use of various cutting and fastening tools. - Conduct a Pair-share Round Robin activity where each trainee identifies a measuring tool and explains its specific	- Assorted electrical tools (hacksaws, stripping knives, side cutters, pliers, tape measure, try square, spirit level, assorted screwdrivers, assorted hammers, multimeters) - Tool safety manuals	Knowledge Checks: -Practical demonstration of tool identification and safe handling. -Oral assessment on tool application. Attitudes: 1. Exhibit respect for	

			3. Select appropriate tools and equipment for specific preparation tasks.	basic checks - Marking tools (pencils, markers) - Care and maintenance of tools	application. - Engage in Q&A Sessions regarding the selection criteria for tools based on the task and material. <u>Follow up Activity:</u> 1. Prepare a checklist of tools required for preparing a 2-meter length of PVC trunking, including their specific uses. (5 Marks) Due date:	- Product specifications for tools	tools and equipment. 2. Demonstrate commitment to safe tool usage. -Oral assessment on tool selection criteria.	
2	1&2	Safety Measures during Trunking Preparation	By the end of the session, the trainee should be able to: 1. Implement	- Understanding	- Engage in a	- OSHA and	Knowledge Checks: -Oral	

			OSHA and EHS safety standards specific to preparing trunking work pieces. 2. Identify potential hazards during measurement, cutting, and handling of trunking materials. 3. Demonstrate the correct use of PPEs for protection against identified hazards.	relevant regulations for material handling - Safety data sheets for materials - Risks associated with cutting tools - Hazards from sharp edges and material dust - Mandatory PPE for preparation tasks (e.g., safety glasses, gloves) - Pre-task safety checks	Think-Pair-Share activity to discuss and apply OSHA/EHS standards to a given preparation scenario. - Participate in a Q&A Sessions to identify and mitigate specific hazards related to cutting and handling trunking materials. - Conduct a Demonstration with Participation where trainees practice donning and doffing PPEs correctly for cutting tasks. <u>Follow up Activity:</u> 1. Create a short incident report describing a	EHS safety manuals - Examples of material safety data sheets (MSDS) - Various PPEs (gloves, safety glasses)	questioning on safety protocols. -Oral questioning on potential hazards. -Practical observation of PPE usage. Attitudes: -Adherence to safety regulations. -Proactive approach to hazard identification.	
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					hypothetical safety breach during trunking preparation and suggest preventive measures. (10 Marks) Due date:			
2	3&4	Measurements for Trunking Work Pieces	By the end of the session, the trainee should be able to: 1. Interpret engineering drawings and work requirements for trunking measurements. 2. Apply various measuring tools to accurately mark out dimensions on trunking	- Reading technical drawings for dimensions - Understanding tolerances and allowances - Using tape measures, rulers, and calipers - Marking methods (pencils, scribes)	- Engage in a Case Study analysis of an engineering drawing to extract specific trunking measurement requirements. - Participate in Demonstrations with Participation on using various measuring tools to accurately mark out	- Engineering drawings/sketches - Measuring tools (tape measures, try squares, spirit levels, rulers) - Trunking	Knowledge Checks: - Practical demonstration of measurement techniques. - Written interpretation of given dimensions.	

			materials. 3. Verify measurement accuracy to prevent material waste and ensure proper fit.	- Double-checking measurements - Importance of precision for subsequent steps	dimensions on trunking samples. - Conduct a Pair-share Round Robin activity where trainees verify each other's marked measurements and provide constructive feedback. <u>Follow up Activity:</u> 1. Measure and mark specific points on a given piece of trunking material according to a provided diagram. (5 Marks) Due date:	samples	-Oral assessment. Attitudes: 1. Commitment to accuracy and precision. 2. Patience in performing detailed tasks.	
	5&6	Cutting Trunking Work Pieces Accurately	By the end of the session, the trainee should be able to: 1. Select	- Hacksaw	- Participate in	- Trunking	Knowledge Checks: -Practical	

			appropriate cutting tools for different trunking materials (PVC, metallic). 2. Apply correct cutting techniques to achieve precise and clean cuts as per measurements. 3. Ensure cut edges are smooth and free from burrs for safe handling and proper assembly.	blades for metallic trunking - Special cutters for PVC trunking - Securing material for cutting - Straight cuts, angle cuts (e.g., 45 degrees for corners) - Deburring tools and methods - Checking for sharp edges	Demonstrations with Participation focusing on the safe and accurate cutting of both PVC and metallic trunking. - Conduct a Peer Teaching session where trainees demonstrate and explain different cutting techniques to their peers. - Engage in a pair-share Round Robin where each trainee inspects and provides feedback on the quality of cut edges made by others.	samples (PVC, metallic) - Cutting tools (hacksaws, trunking cutters, files, deburring tools) - PPEs (gloves, safety glasses)	assessment of cutting accuracy. -Oral questioning on tool selection for specific materials. -Oral questioning Attitudes: -Diligence in achieving clean and precise cuts. -Awareness of material characteristics during cutting.	
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					<u>Follow up Activity:</u> 1. Cut a 50cm length of PVC trunking and a 30cm length of metallic trunking, ensuring smooth and burr-free edges. (15 Marks) Due date:			
2	7&8	Housekeeping and Tool Storage	By the end of the session, the trainee should be able to: 1. Explain the importance of housekeeping in an electrical workshop environment. 2. Apply systematic procedures for cleaning the	- Safety implications (preventing trips, hazards) - Efficiency and productivity benefits - Waste disposal (material scraps, dust) - Cleaning work	- Engage in a Group Discussion about the direct link between good housekeeping and workplace safety/efficiency. - Participate in Demonstrations with Participation on	- Cleaning supplies (brooms, dustpans, waste bins) - Tool storage solutions (toolboxes, racks) - Workshop layout diagrams	Knowledge Checks: -Oral questioning on housekeeping benefits. -Practical observation of workstation cleanup.	

			workplace after trunking preparation.	surfaces and floors	proper waste disposal and cleaning up after a preparation task.			
			3. Demonstrate proper storage techniques for tools and equipment.	- Organizing tools (toolboxes, racks) - Maintenance before storage (cleaning, oiling)	- Conduct a Gallery Walk of a well-organized and a poorly organized tool storage area, discussing the pros and cons. <u>Follow up Activity:</u> 1. Develop a short daily checklist for housekeeping activities in a trunking preparation workstation. (5 Marks) Due date:		Oral questioning on storage techniques for tools and equipment. Attitudes: -Responsibility for maintaining a clean and safe environment. -Discipline in organizing tools.	
3	1&2	Marking Mounting Points for Trunking Systems	By the end of the session, the trainee should be able to:				Knowledge Checks:	

			1. Interpret installation plans to identify optimal trunking routes and mounting point locations. 2. Apply IEC standards for spacing and alignment when marking mounting points. 3. Utilize appropriate marking tools to accurately transfer measurements to the mounting surface.	- Reading layout drawings for trunking paths - Understanding architectural features - Spacing requirements for supports - Alignment principles (horizontal, vertical) - Using spirit levels, chalk lines, rulers, and pencils - Ensuring marks are clear and precise	- Engage in a Case Study where trainees analyze an installation blueprint and propose ideal trunking routes and mounting points. - Participate in a Demonstrations with Participation session on using spirit levels and chalk lines to accurately mark out a horizontal trunking path on a mock-up wall. - Conduct a Think-Pair-Share activity to discuss the importance of accurate marking for structural integrity and aesthetics.	- Mock-up wall sections - Installation blueprints/diagrams - Marking tools (spirit levels, chalk lines, rulers, pencils) - IEC standards (relevant sections on mounting)	-Practical assessment of marked points accuracy. -Oral questioning on IEC standards for mounting. -Oral questioning Attitudes: 1. Commitment to precision in marking. 2. Appreciation	
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					<u>Follow up Activity:</u> 1. Given a mock-up wall and a simple trunking layout, accurately mark all required mounting points according to provided specifications. (10 Marks) Due date:		for adherence to standards.	
3	3&4	Creating Trunking Angles	By the end of the session, the trainee should be able to: 1. Explain the purpose and application of various trunking angles (e.g., 45°, 90°). 2. Measure and mark angles accurately on trunking work pieces for neat	- Aesthetics and professional finish - Proper routing around obstacles - Using protractors, combination squares - Techniques for	- Engage in a Brainstorming session to identify scenarios where 45° and 90° trunking angles are essential. - Participate in Demonstrations with Participation on how to	- Trunking samples - Measuring tools (protractors, combination squares, rules) - Cutting tools (hacksaws, miter boxes) - Files and deburring tools	Knowledge Checks: -Practical assessment of angle cutting accuracy. -Oral questioning on angle applications.	

			corners and bends.	45 and 90-degree angles	accurately measure, mark, and cut trunking at 45° and 90° angles using hand tools.			
			3. Utilize appropriate cutting and forming techniques to create precise trunking angles.	- Specific tools for angle cuts - Deburring and joining angled pieces	- Conduct a Peer Teaching activity where trainees instruct each other on deburring and preparing angled trunking pieces for joining. <u>Follow up Activity:</u> 1. Cut and prepare two pieces of trunking to form a perfect 90-degree external corner, ensuring neat joints. (15 Marks) Due date:		- Practical demonstration Attitudes: -Diligence in achieving professional finishes. -Creative problem-solving for complex bends.	
	5&6	Basic Mounting of	By the end of the session, the				Knowledge Checks:	

		PVC Trunking	trainee should be able to: 1. Prepare mounting surfaces and trunking pieces for installation. 2. Apply basic fastening techniques to secure PVC trunking as per marked points. 3. Ensure trunking is level, straight, and securely fixed according to IEC standards.	- Cleaning surfaces, pre-drilling holes - Ensuring trunking pieces are ready for mounting - Using screws, wall plugs, or adhesives for PVC - Correct torque and spacing - Checking for levelness and alignment with a spirit level - Confirming secure attachment	- Engage in Demonstrations with Participation on preparing a mock-up wall and a PVC trunking piece for mounting. - Participate in a Round Robin activity where trainees take turns securing sections of PVC trunking to a practice board, receiving peer feedback. - Conduct a Q&A Sessions to address common challenges in PVC trunking mounting and how to overcome them while	- Mock-up walls/practice boards - PVC trunking sections - Fastening tools and hardware (drills, screwdrivers, screws, wall plugs) - Spirit levels - IEC standards (relevant sections on mounting)	- Practical assessment of PVC trunking installation. - Oral questioning on fastening methods. - Oral questioning Attitudes: 1. Commitment to quality craftsmanship. 2. Patience and	
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					adhering to IEC standards. <u>Follow up Activity:</u> 1. Mount a 1-meter length of PVC trunking onto a vertical surface, ensuring it is perfectly level and securely fixed. (20 Marks) Due date:		attention to detail during installation.	
3	7&8	Continuous Assessment Test (CAT) 1	By the end of the session, the trainee should be able to: a. Demonstrate comprehensive theoretical understanding of identifying trunking components and preparing work pieces. b. Execute practical tasks	ASSESSMENT COVERAGE - Workplace safety and PPEs - Trunking types, sizes, and accessories - Electrical tools and equipment for preparation - Applying	- Engage in a structured written examination covering theoretical aspects of LO1 and LO2. - Participate in practical stations to demonstrate identification of components, tools, and basic	- Examination papers - Practical assessment sheets - Relevant tools, trunking samples, and accessories for practical stations	Knowledge Checks: 1. Comprehensive written test. 2. Practical skills assessment. Attitudes: 1. Honesty and integrity during assessment. 2. Self-discipline in time	

			related to trunking identification, measurement, and cutting with proficiency. c. Apply safety protocols and housekeeping procedures effectively during practical assessment.	safety measures, measurements, and cutting techniques - Housekeeping activities TEST FORMAT - Written questions (multiple choice, short answer) - Practical identification and demonstration tasks INSTRUCTIONS - Review of test rules and time allocation - Clarification of assessment criteria	measurement/cutting skills. - Adhere to all assessment instructions and time limits. Follow up Activity: 1. Self-reflection on performance during CAT 1, identifying areas for improvement. (Not Graded) Due date:		management.	
4	1&2	Mounting Metallic	By the end of the session, the				Knowledge Checks:	

		Trunking	trainee should be able to: 1. Differentiate mounting considerations for metallic trunking compared to PVC. 2. Apply appropriate fastening techniques for securing metallic trunking. 3. Ensure continuity of earthing for metallic trunking systems as per safety standards.	- Heavier weight, requiring robust fixings - Earthing requirements for safety - Screws into masonry, toggle anchors for hollow walls - Using metallic clips or saddles - Importance of proper bonding - Testing for electrical continuity	- Engage in a Give One Take One activity to share insights on the challenges and solutions for mounting metallic trunking. - Participate in Demonstrations with Participation on securing metallic trunking to a mock-up surface using various fasteners. - Conduct a Q&A Sessions to discuss earthing requirements for metallic trunking installations. <u>Follow up</u>	- Metallic trunking sections - Fastening hardware for metallic trunking - Drills, screwdrivers, spirit levels - Continuity tester	- Practical assessment of metallic trunking mounting. - Practical Demonstrations - Oral questioning on earthing principles. Attitudes: - Meticulousness in safety	
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					<u>Activity:</u> 1. Mount a 1-meter length of metallic trunking onto a mock-up surface, including necessary earthing connections. (20 Marks) Due date:		procedures. -Responsibility for structural integrity.	
4	3&4	Mounting Galvanized Iron Trunking	By the end of the session, the trainee should be able to: 1. Identify specific handling and mounting requirements for galvanized iron trunking. 2. Apply appropriate techniques for cutting and joining	- Corrosion resistance, but susceptible to damage during cutting - Heavier and more rigid than PVC or standard metallic - Specialized tools for GI cutting (e.g., grinders with appropriate	- Engage in a Brainstorming session on the unique challenges presented by galvanized iron trunking installation. - Participate in Demonstrations with Participation on safely cutting	- Galvanized iron trunking sections - Specialized cutting tools (grinders, metal snips)	Knowledge Checks: -Practical assessment of GI trunking handling. -Oral questioning on material-specific techniques.	

			galvanized iron trunking. 3. Secure galvanized iron trunking to various surfaces, considering its rigidity and weight.	blades) - Methods for achieving neat joints - Stronger anchors and supports - Ensuring proper alignment and support over lengths	and joining galvanized iron trunking using specialized tools. - Conduct a Group Discussion to evaluate the best mounting strategies for different surface types when working with GI trunking. <u>Follow up Activity:</u> 1. Prepare and mount a 0.5-meter section of galvanized iron trunking, demonstrating proper cutting, joining, and fastening. (20 Marks) Due date:	- Heavy-duty fasteners - PPE (face shield, hearing protection)	-Oral questioning Attitudes: -Caution and precision in handling heavy materials. -Adaptability to different material properties.	
	5&6	Advanced Mounting	By the end of the session, the				Knowledge Checks:	

		Techniques and Support Systems	trainee should be able to: 1. Install various support systems for trunking in challenging environments (e.g., ceilings, uneven walls). 2. Apply techniques for supporting trunking runs over long distances and around complex obstacles. 3. Troubleshoot common mounting issues and	- Suspended rod systems, brackets, trapeze supports - Load bearing calculations (basic) - Negotiating corners, columns, and ceiling obstructions - Expansion joints for long runs - Dealing with misaligned holes, loose fixings - Rectifying	- Engage in a Case Study analysis of complex installation scenarios, proposing suitable support systems and routing strategies. - Participate in Demonstrations with Participation on installing different types of support brackets and suspended systems for trunking. - Conduct a Role Play exercise where trainees simulate troubleshooting	- Various trunking support brackets and rods - Mock-up ceiling/complex wall sections - Fastening tools and hardware - Case study examples of complex installations	-Practical demonstration of advanced mounting. -Written problem-solving exercise. -Oral questioning Attitudes:	
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			implement corrective actions.	aesthetic imperfections	common mounting problems and present solutions. <u>Follow up Activity:</u> 1. Design a support system for a 5-meter trunking run across a ceiling, detailing the type of supports and spacing. (15 Marks) Due date:		1. Innovative thinking in challenging situations. 2. Perseverance in achieving difficult installations.	
4	7&8	Housekeeping Activities After Mounting	By the end of the session, the trainee should be able to: 1. Explain the importance of comprehensive cleanup after trunking mounting.	- Removing all debris (cut-offs, dust, packaging) - Ensuring area is free of tripping hazards	- Engage in a Group Discussion about the environmental impact of improper waste disposal in electrical	- Cleaning supplies - Waste bins (labeled for segregation) - Sample waste materials from	Knowledge Checks: -Oral questioning on waste management protocols.	

			2. Implement effective waste disposal and material segregation practices. 3. Conduct a final check of the work area for safety and cleanliness.	- Segregation of recyclable materials (metal, plastic) - Proper disposal of non-recyclable waste - Inspecting for sharp edges, protruding fasteners - Ensuring tools are accounted for and stored	installations. - Participate in Demonstrations with Participation on systematically cleaning a work area after mounting, emphasizing waste segregation. - Conduct a Gallery Walk of mock-up completed installation sites, identifying good and bad housekeeping practices. <u>Follow up Activity:</u> 1. Lead a team in a simulated post-	mounting	-Practical observation of cleanup efforts. -Practical observation Attitudes: -Environmental consciousness. - Responsibility for a safe working environment.	
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					mounting cleanup, ensuring all waste is properly segregated and the area is safe. (10 Marks) Due date:			
5	1&2	Electrical Cables	By the end of the session, the trainee should be able to: 1. Identify standard electrical cable colour codes as per IEC standards for various phases, neutral, and earth. 2. Differentiate between various electrical cable sizes (e.g., 1.0mm², 1.5mm², 2.5mm²,	- Live (phase) colors (brown, black, grey) - Neutral colour (blue) - Earth colour (green and yellow) - Cross-sectional area (mm²) and current carrying capacity - Factors affecting cable sizing (e.g.,	- Engage in a Think-Pair-Share activity to recall and discuss the significance of various cable colour codes. - Participate in a Demonstrations with Participation session on accurately measuring cable cross-sectional	- Various electrical cable samples (different colours and sizes) - IEC wiring regulations chart (excerpts) - Multimeter (for basic resistance check) - Cable charts for current ratings	Knowledge Checks: -Practical identification of cable colour codes and sizes. -Written quiz on cable selection principles.	

			4.0mm², 6.0mm²) and their applications. 3. Select appropriate cable types and sizes for different electrical loads and circuit requirements.	voltage drop, protection) - Matching cable size to load requirements - Adherence to wiring regulations	area and identifying different cable sizes. - Conduct a Case Study analysis where trainees select correct cable sizes and colours for various hypothetical circuit diagrams. <u>Follow up Activity:</u> 1. Label 5 different cable samples with their correct colour code, size, and typical application. (10 Marks) Due date:		-Practical identification Attitudes: -Attention to detail in cable identification. -Responsibility for safety through correct wiring.	
5	3&4	Cable Laying Techniques in Trunking (Single	By the end of the session, the trainee should be able to: 1. Apply safe	- Pulling	- Engage in a	- Trunking	Knowledge Checks: -Practical	

		Core)	and efficient methods for laying single core cables within trunking systems. 2. Ensure proper dressing and routing of cables to avoid damage and facilitate future maintenance. 3. Adhere to maximum fill ratios and bending radii for cables within trunking as per IEC standards.	techniques without kinking or stressing - Use of cable lubricant (if necessary) - Neat arrangement within trunking - Avoiding overcrowding - Understanding trunking fill capacity limits - Observing minimum bending radii to prevent insulation damage	Demonstrations with Participation session on proper cable pulling and laying techniques for single core cables in a mock trunking run. - Participate in a Round Robin activity where each trainee practices dressing and routing cables neatly within a trunking section. - Conduct a Q&A Sessions to discuss common mistakes in cable laying and how they can lead to safety hazards or system failures. <u>Follow up</u>	sections (mounted) - Single core electrical cables - Cable pulling aids (if applicable) - IEC wiring standards (relevant sections on cable installation)	assessment of cable laying neatness and technique. -Oral questioning on fill ratios and bending radii. -Oral questioning Attitudes: 1. Commitment to neatness and organization. 2. Awareness of cable integrity during installation.	
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					<u>Activity:</u> 1. Lay 3 single core cables (different colours) neatly within a 1-meter trunking section, demonstrating correct routing and dressing. (15 Marks) Due date:			
	5&6	Installation of Single-Phase Intake Point Components (Part 1)	By the end of the session, the trainee should be able to: 1. Identify the components of a single-phase electrical intake point. 2. Explain the function and safety considerations for installing 1-	- Role of the intake point in a domestic/light commercial setup - Main components (cut-out, meter, CCU) - Function as main protective device - Safety procedures for	- Engage in a Group Discussion to map out the typical layout and function of a single-phase electrical intake point. - Participate in Demonstrations with Participation on the safe handling	- Mock-up intake board with trunking - 1-phase cut-out units - Wiring diagrams for single-phase intake - Insulated tools	Knowledge Checks: -Practical assessment of cut-out installation. -Oral questioning on safety protocols for intake points. -Practical Demonstrations -Oral questioning on	

			phase cut-outs. 3. Install single-phase cut-outs correctly within a trunking-based intake system.	installation (isolation) - Mounting cut-out enclosure in trunking - Wiring connections as per diagram	and installation of a 1-phase cut-out within a mock intake setup. - Conduct a Think-Pair-Share activity to brainstorm potential risks during cut-out installation and how to mitigate them. <u>Follow up Activity:</u> 1. Install a 1-phase cut-out into a mock trunking intake board, ensuring all connections are secure and polarity is correct. (20 Marks) Due date:		safety protocols for intake points. -Oral questioning Attitudes: -Responsibility for precise and safe wiring. -Adherence to utility company requirements. -Practical Demonstrations	
5	7&8	Installing Single-Phase	By the end of the session, the				Knowledge Checks:	

		Intake Point Components (Part 2)	trainee should be able to: 1. Describe the operation and installation requirements for single-phase energy meters. 2. Explain the role and components of a Consumer Control Unit (CCU) in a single-phase system. 3. Install energy meters and CCUs as part of a complete trunking-based intake point.	- Principles of energy measurement - Correct wiring sequence for connection - Main switch, RCDs/RCBOs, MCBs - Importance of correct circuit allocation - Combining cut-out, meter, and CCU - Ensuring neatness and accessibility	- Engage in a KWL activity to identify prior knowledge, what they want to learn, and what they learned about energy meters. - Participate in Demonstrations with Participation on the internal wiring and connection of a single-phase CCU. - Conduct a Peer Teaching session where trainees demonstrate the complete assembly of a single-phase intake point with all components.	- Mock-up intake board - 1-phase energy meters - Consumer Control Units (CCU) with various breakers - Wiring tools and diagrams	-Practical assessment of energy meter and CCU wiring. -Oral questioning on circuit protection principles. -Practical Demonstrations Attitudes: 1. Diligence in conforming to wiring	
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					<u>Follow up Activity:</u> 1. Wire a complete single-phase intake point on a mock board, incorporating a cut-out, energy meter, and a basic CCU. (25 Marks) Due date:		standards. 2. Foresight in planning for future maintenance.	
6	1&2	Earth Rods and Switch Fuse Installation	By the end of the session, the trainee should be able to: 1. Explain the purpose and importance of earthing systems, including earth rods. 2. Describe the function and installation	- Protection against electric shock - Role of the earth rod in dissipating fault current - Combined isolation and overcurrent	- Engage in a Group Discussion on the critical importance of proper earthing in preventing electrical accidents. - Participate in Demonstrations with	- Earth rod (sample) and clamps - Single-phase switch fuse units - Wiring diagrams showing earthing and fuse	Knowledge Checks: -Oral questioning on earthing principles and switch fuse function. -Practical demonstration of switch fuse	

			procedures for single-phase switch fuses. 3. Install earth rods and connect single-phase switch fuses correctly as part of an electrical installation.	protection - Correct rating and placement - Driving and connecting earth rods (simulated) - Wiring and securing switch fuses	Participation on the simulated installation of an earth rod and its connection to the main earthing terminal. - Conduct a Think-Pair-Share activity to discuss the various applications and safety features of a single-phase switch fuse. <u>Follow up Activity:</u> 1. Draw a schematic diagram showing the connection of an earth rod to a single-phase intake point via a switch fuse. (10 Marks) Due date:	connections - Safety guidelines for earthing	wiring. -Practical demonstration Attitudes: -Rigorous adherence to safety standards. -Responsibility for protecting human life.	
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6	3&4	Lighting Circuit Installation - One Way Control	By the end of the session, the trainee should be able to: 1. Identify the components and wiring requirements for a one-way lighting circuit. 2. Interpret wiring diagrams for single-switch, one-way lighting control. 3. Install a one-way lighting circuit within a trunking system,	- Live, switched live, neutral, earth connections - Single switch control for a light point - Understanding circuit paths - Colour code application - Mounting switch and lamp holder in trunking - Connecting	- Engage in a KWL activity to outline existing knowledge, desired learning, and what was learned about one-way lighting circuits. - Participate in Demonstrations with Participation on interpreting a one-way lighting circuit diagram and making connections in trunking. - Conduct a Peer Teaching session where trainees guide each other through the step-	- Mock-up board with trunking - Single-pole switches, lamp holders, light bulbs - Electrical cables (various colours) - Wiring diagrams	Knowledge Checks: -Practical assessment of one-way lighting circuit installation and testing. -Oral questioning on circuit operation. -Practical Demonstration Attitudes: -Attention to	
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			ensuring functional operation.	wires securely - Testing circuit for functionality	by-step installation of a one-way lighting circuit on a mock-up. <u>Follow up Activity:</u> 1. Install and test a functional one-way lighting circuit on a mock-up board, completely housed in trunking. (25 Marks) Due date:		detail in wiring connections. -Problem-solving approach to circuit faults.	
	5&6	Lighting Circuit Installation - Two Way Control	By the end of the session, the trainee should be able to: 1. Explain the operational principle of a two-way lighting circuit.	- Use of two two-way switches - Strapping wires (common, L1, L2)	- Engage in a Brainstorming session on real-world applications where two-way lighting control is necessary.	- Mock-up board with trunking - Two-way switches, lamp holders, light bulbs - Electrical cables (various colours)	Knowledge Checks: -Practical assessment of two-way lighting circuit installation and testing.	

			2. Interpret complex wiring diagrams for two-switch, two-way lighting control. 3. Install a two-way lighting circuit within a trunking system, ensuring independent control from two locations.	- Tracing paths for two-way switching - Ensuring correct connections - Mounting two-way switches and lamp holder - Connecting strappers and commons - Testing for independent operation	- Participate in Demonstrations with Participation on deciphering a two-way lighting circuit diagram and performing the intricate wiring. - Conduct a Round Robin where trainees practice wiring specific sections of a two-way circuit, then rotate to complete the circuit and test it. <u>Follow up Activity:</u> 1. Install and test a functional two-way lighting circuit on a mock-up board, ensuring both switches control	- Wiring diagrams for two-way switching	-Oral questioning on switching principles. -Practical Demonstration Attitudes: -Precision in complex wiring. -Perseverance in troubleshooting advanced circuits.	
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					the light independently. (30 Marks) Due date:			
6	7&8	Lighting Circuit Installation - Intermediate Control	By the end of the session, the trainee should be able to: 1. Explain the operational principle of an intermediate lighting circuit. 2. Interpret wiring diagrams for intermediate switching (control from three or more locations). 3. Install an	- Combination of two-way switches and an intermediate switch - Controlling a light from three or more points - Understanding the switching matrix - Correct placement of intermediate switch - Mounting	- Engage in a Case Study of a building requiring multi-location lighting control, discussing the role of intermediate switching. - Participate in Demonstrations with Participation on the advanced wiring required for an intermediate lighting circuit in trunking. - Conduct a	- Mock-up board with trunking - Two-way switches, intermediate switches, lamp holders, light bulbs - Electrical cables (various colours) - Advanced wiring diagrams for intermediate switching	Knowledge Checks: -Practical assessment of intermediate lighting circuit installation and testing. -Oral questioning on multi-point control logic. -Oral	

			intermediate lighting circuit within a trunking system, demonstrating multi-location control.	switches (two-way and intermediate) and lamp holder - Connecting cross-over wires - Thorough testing for all control points	Group Discussion on common errors encountered during intermediate circuit wiring and strategies for debugging. <u>Follow up Activity:</u> 1. Install and test a functional intermediate lighting circuit (three control points) on a mock-up board, fully trunked. (35 Marks) Due date:		questioning Attitudes: 1. Meticulous attention to complex wiring. 2. High standard of professional workmanship.	
7	1&2	Looping-in Methods at Ceiling Rose and Joint Boxes	By the end of the session, the trainee should be able to: 1. Explain the concept and advantages of looping-in wiring	- Reducing terminal connections at switches - Centralizing	- Engage in a Think-Pair-Share activity to compare looping-in	- Mock-up ceiling sections with	Knowledge Checks: -Practical assessment of looping-in connections.	

			methods. 2. Apply looping-in techniques for connections at ceiling roses. 3. Implement looping-in at joint boxes for circuit extensions and taps.	connections at lighting points/boxes - Connecting live, neutral, earth, and switch wire at the rose - Ensuring correct polarity - Using joint boxes for branching circuits - Maintaining accessibility for future work	methods with traditional wiring approaches and discuss their benefits. - Participate in Demonstrations with Participation on how to correctly perform looping-in connections within a ceiling rose. - Conduct a Round Robin session where trainees practice making looping-in connections in different types of joint boxes. <u>Follow up Activity:</u> 1. Wire a lighting point using the looping-in	ceiling roses - Various joint boxes - Electrical cables, lamp holders, light bulbs - Wiring diagrams specific to looping-in	-Oral questioning on circuit integrity. -Practical Demonstration Attitudes: -Precision in small wiring components. -Understanding of circuit architecture.	
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					method at a ceiling rose, ensuring correct connections and functionality. (20 Marks) Due date:			
7	3&4	Looping-in Methods at Switches	By the end of the session, the trainee should be able to: 1. Describe the application of looping-in methods directly at switches. 2. Interpret wiring diagrams illustrating looping-in at switch terminals.	- Bringing live and neutral directly to the switch box - Advantages for certain installations - Connecting multiple live and neutral conductors - Ensuring sufficient terminal capacity	- Engage in a Case Study focusing on situations where looping-in at switches is the preferred method. - Participate in Demonstrations with Participation on safely and correctly performing looping-in connections at a switch terminal.	- Mock-up wall sections with trunking and switch boxes - Two-gang switches, electrical cables - Wiring diagrams illustrating switch looping-in	Knowledge Checks: -Practical assessment of switch looping-in wiring. -Oral questioning on alternative wiring strategies.	

			3. Install wiring using looping-in technique directly into switch boxes within a trunking system.	- Mounting switch boxes - Making secure and neat connections - Adherence to enclosure fill limits	- Conduct a Peer Teaching session where trainees explain and demonstrate the process of terminating looped-in cables at a switch. <u>Follow up Activity:</u> 1. Wire a two-gang switch using the looping-in method, ensuring proper termination of all conductors within a trunked switch box. (20 Marks) Due date:		-Oral questioning Attitudes: -Adaptability to different wiring solutions. -Attention to detail in confined spaces.	
	5&6	Power Circuits - Socket Outlets (Ring and	By the end of the session, the trainee should be able to: 1. Differentiate	- Characteristics,	- Engage in a		Knowledge Checks:	

		Radial)	between ring final circuits and radial final circuits for socket outlets. 2. Interpret wiring diagrams for both ring and radial socket outlet circuits. 3. Install socket outlets within a trunking system, configured as either a ring or a radial circuit.	advantages, and disadvantages of each - Typical applications - Correct termination at socket terminals - Continuity of earth and main conductors - Routing cables to multiple socket points - Ensuring sufficient space and security	KWL activity to explore the characteristics, pros, and cons of ring versus radial circuits. - Participate in Demonstrations with Participation on wiring a single socket outlet correctly, then extending it for a radial circuit. - Conduct a Group Discussion on how to plan and lay out a ring final circuit within a given floor plan using trunking. <u>Follow up Activity:</u> 1. Install three 13A socket outlets as part of	- Mock-up board with trunking - 13A socket outlets, electrical cables - Wiring diagrams for ring and radial circuits - Circuit testers	-Practical assessment of socket outlet installation. -Oral questioning on ring and radial circuit principles. -Practical assessment of socket outlet installation. Attitudes: -Thoroughness in wiring power circuits. -Understanding of circuit loading and protection.	
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					a radial circuit within a trunking system on a mock-up board. (25 Marks) Due date:			
7	7&8	Power Circuits - Cooker Units and Water Heaters	By the end of the session, the trainee should be able to: 1. Explain the specific electrical requirements and safety precautions for cooker units. 2. Describe the installation procedures for connecting water heaters.	- High current demand, dedicated circuit - Correct cable size and circuit breaker rating - Double pole isolation - Similar high demand considerations - Use of appropriate isolators (e.g., fused spur)	- Engage in a Brainstorming session on the unique safety challenges and electrical demands of cooker and water heater installations. - Participate in Demonstrations with Participation on the safe and correct wiring of a cooker control unit to a simulated supply.	- Mock-up board with trunking - Cooker control units, water heater isolators - High-current cables, appropriate circuit breakers - Wiring regulations for high-power appliances	Knowledge Checks: -Practical assessment of high-power appliance wiring. -Oral questioning on load calculation and circuit protection.	

			3. Install cooker units and water heaters, ensuring appropriate cable sizing and protective devices.	- Adherence to regulations for high-power appliances - Importance of proper earthing	- Conduct a Case Study analysis of common installation mistakes for these high-power appliances and their consequences. <u>Follow up Activity:</u> 1. Wire a cooker control unit and a water heater isolator on a mock-up board, ensuring correct cable and fuse ratings. (30 Marks) Due date:		-Oral questioning Attitudes: 1. Utmost diligence in high-risk installations. 2. Commitment to client safety.	
8	1&2	Housekeeping after Cable Installation	By the end of the session, the trainee should be able to: 1. Explain the importance of maintaining cleanliness during and	- Managing cable offcuts, insulation scraps, and packaging - Preventing	- Engage in a Group Discussion about the aesthetic and safety benefits of	- Cleaning	Knowledge Checks: -Oral questioning on waste management best practices.	

			after cable installation. 2. Apply effective methods for managing cable waste and debris. 3. Conduct a final inspection of the work area to ensure safety and neatness.	accumulation of debris - Distinguishing copper, PVC, and other materials for recycling/disposal - Use of designated waste bins - Ensuring the area is clear for subsequent trades or client inspection - Tool accountability	meticulous housekeeping in a partially completed installation. - Participate in Demonstrations with Participation on efficient techniques for collecting and segregating cable waste during a simulated installation. - Conduct a Gallery Walk through a mock installation, identifying areas that require improved housekeeping and suggesting solutions. <u>Follow up Activity:</u>	supplies, waste bins - Various cable offcuts and packaging - Checklist for post-installation cleanup	-Practical observation of cleanup and organization. -Practical observation -Oral questioning Attitudes: -Professional pride in a tidy work area. -Respect for environmental guidelines.	
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					1. Perform a complete cleanup of a mock cable installation workstation, ensuring all tools are returned and waste is segregated. (10 Marks) Due date:			
8	3&4	Inspection and testing of Trunking Installations	By the end of the session, the trainee should be able to: 1. Explain the purpose and scope of visual inspection in electrical installations. 2. Identify common defects and non-conformities through systematic	- Checking for physical damage, correct component use, proper routing - Ensuring compliance with standards before testing - Poorly secured trunking, untidy wiring, incorrect accessories - Damage to insulation, missing	- Engage in a Case Study where trainees are given images of faulty installations and asked to identify visual defects. - Participate in a Demonstrations with Participation session on conducting a systematic visual	- Mock-up trunking installations (some with deliberate faults) - Visual inspection checklists - Relevant sections of IEC standards	Knowledge Checks: -Practical assessment of fault identification. -Oral questioning on inspection criteria.	

			visual checks. 3. Document visual inspection findings accurately and suggest corrective actions.	protective covers - Using inspection checklists - Reporting non-conformities	inspection of a completed trunking installation mock-up. - Conduct a Round Robin activity where trainees perform a visual inspection of each other's work and provide feedback using a checklist. <u>Follow up Activity:</u> 1. Conduct a visual inspection of a simulated trunking installation (with deliberate faults) and document 5 identified non-conformities. (15 Marks) Due date:		-Oral questioning Attitudes: 1. Diligence and attention to detail. 2. Commitment to quality assurance.	
	5&6	Continuity test	By the end of the session, the					

		Procedures	trainee should be able to: 1. Explain the purpose of continuity testing in electrical circuits. 2. Describe the procedures for performing continuity tests on protective conductors and ring final circuits. 3. Conduct continuity tests using appropriate test equipment as per IEC standards.	- Ensuring unbroken protective conductor paths - Verifying proper connections in circuits - Use of low-resistance ohmmeter - Testing earth continuity, equipotential bonding - Ring final circuit continuity (end-to-end resistance) - Calibration and safe operation of continuity testers - Interpreting test results	- Engage in a Think-Pair-Share activity to discuss why continuity is crucial for electrical safety and circuit function. - Participate in Demonstrations with Participation on setting up and performing continuity tests on mock protective conductors and a ring circuit. - Conduct a Peer Teaching session where trainees demonstrate and explain the correct interpretation of	- Mock-up trunking installation with earth conductors and ring circuit - Continuity tester (low resistance ohmmeter) - IEC standards (relevant sections on testing) - Test result recording sheets	-Practical assessment of continuity testing. -Oral questioning on test result interpretation. -Practical Demonstration Attitudes: -Responsibility for safety through testing.	
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					continuity test readings. <u>Follow up Activity:</u> 1. Perform continuity tests on the protective conductors and a simulated ring final circuit on a mock-up board, recording all readings. (20 Marks) Due date:		-Precision in using test equipment.	
8	7&8	Continuous Assessment Test (CAT) 2	By the end of the session, the trainee should be able to: 1. Demonstrate proficiency in mounting various trunking types and implementing support systems. 2. Execute complex cable	- Mounting trunking work pieces (all types, angles, supports) - Cable colour codes, sizes, and laying techniques - Single-phase intake point	- Engage in practical installation tasks covering various aspects of LO3 and LO4. - Complete a written section	- Practical assessment stations with required materials and tools - Written examination papers	Knowledge Checks: -Practical skills assessment on complex installations. 2. Written comprehensive test. Attitudes: 1. Professionalism under pressure. 2. Integrity in	

			installations including intake points, lighting, and power circuits. 3. Conduct visual inspections and perform basic electrical tests (e.g., continuity) accurately.	components (cut-outs, meters, CCUs) - Earthing and switch fuse installation - Lighting circuits (one-way, two-way, intermediate) - Looping-in methods - Power circuits (socket outlets, cooker, water heater) - Housekeeping after cable installation - Visual inspection of installations - Continuity test procedures TEST FORMAT - Comprehensive practical examination (installation tasks)	addressing theoretical knowledge and problem-solving scenarios related to the covered topics. - Maintain focus and adhere to all assessment rules. Follow up Activity: 1. Review CAT 2 feedback and revise areas of weakness identified in the assessment. (Not Graded) Due date:	- Safety equipment	demonstrating competence.	
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				- Written questions on principles and troubleshooting INSTRUCTIONS - Strict adherence to safety protocols during practical tasks - Time management for multi-part assessment				
9	1&2	Polarity Test Procedures	By the end of the session, the trainee should be able to: 1. Explain the fundamental importance of correct polarity in electrical installations.	- Ensuring protective devices operate correctly - Preventing shock hazards	- Engage in a Group Discussion to illustrate real-world scenarios where incorrect polarity could lead to dangerous situations.	- Mock-up trunking installation with various accessories - Polarity tester, voltage	Knowledge Checks: -Practical assessment of polarity testing.	

			2. Describe the procedures for performing polarity tests on accessories and circuits.	- Verifying correct connection of live, neutral, and earth at accessories - Using a voltage tester or approved polarity test equipment	- Participate in Demonstrations with Participation on how to safely and accurately perform polarity tests at various socket outlets and switches.	indicator - IEC standards (relevant sections on testing) - Test result recording sheets	-Oral questioning on safety implications of incorrect polarity.	
			3. Conduct polarity tests using appropriate test equipment as per IEC standards.	- Safe handling of test equipment - Interpreting correct and incorrect polarity readings	- Conduct a Round Robin activity where trainees practice polarity testing on mock circuits with deliberate polarity faults, identifying and correcting them. <u>Follow up Activity:</u> 1. Perform polarity tests on five different accessories (sockets,		-Practical assessment of polarity testing. Attitudes: - Meticulousness in safety-critical checks. -Responsibility for correct circuit configuration.	

					switches, lamp holders) on a mock-up board, recording results. (15 Marks) Due date:			
9	3&4	Ring Test Procedures	By the end of the session, the trainee should be able to: 1. Explain the specific purpose of the ring continuity test for ring final circuits. 2. Describe the procedures for performing ring final circuit tests (R1+R2, Rn, R1+Rn).	- Verifying the integrity of both live and neutral conductors in a ring - Confirming earth continuity throughout the ring - Measuring R1+R2 at each socket outlet - End-to-end resistance measurements	- Engage in a Case Study of a faulty ring final circuit, using theoretical test results to diagnose the problem. - Participate in Demonstrations with Participation on systematically performing R1+R2, R1, and Rn tests on a mock ring final circuit.	- Mock-up ring final circuit (fully wired) - Multi-function electrical tester or separate continuity/insulation resistance testers - IEC standards (relevant sections on testing) - Ring circuit test result	Knowledge Checks: -Written interpretation of ring test results. -Practical assessment of ring circuit testing.	

			3. Conduct ring final circuit tests using appropriate test equipment to confirm circuit integrity.	- Use of multi-function tester or individual testers - Comparing readings to expected values and regulations	- Conduct a Peer Teaching session where trainees explain the significance of varying test results for ring circuits. <u>Follow up Activity:</u> 1. Perform a complete set of ring final circuit tests on a mock-up ring circuit, accurately recording all results and identifying any non-conformities. (20 Marks) Due date:	sheet	-Oral questioning Attitudes: 1. Analytical approach to diagnostics. 2. Commitment to thorough testing.	
	5&6	Ethical Work Practices - Personal Management	By the end of the session, the trainee should be able to: 1. Explain the	- Understanding	- Engage in a		Knowledge Checks: -Written	

			components of personal management (self-awareness, self-esteem, emotional intelligence, stress management, assertiveness). 2. Identify scenarios where strong personal management skills contribute to professional success in electrical work. 3. Develop strategies for improving self-awareness, managing stress, and practicing assertiveness in	personal strengths and weaknesses - Building confidence in professional abilities - Recognizing and managing one's own emotions - Empathy towards colleagues and clients - Techniques for coping with work-related stress -Communicating needs and boundaries professionally	KWL activity on personal management, identifying prior knowledge and learning goals. - Participate in a Role Play scenario to practice assertive communication in a workplace conflict situation. - Conduct a Group Discussion on how emotional intelligence impacts teamwork and client relations in	- Handouts on emotional intelligence and assertiveness - Case studies of workplace interpersonal challenges	reflection on self-awareness. -Oral discussion on personal management concepts. -Oral questioning Attitudes: -Openness to self-improvement.	
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			the workplace.		technical roles. <u>Follow up Activity:</u> 1. Write a short reflection on a personal strength and a weakness related to workplace interaction, and outline a plan for improvement. (10 Marks) Due date:		-Empathy towards others.	
9	7&8	Ethical Work Practices - Policies and Guidelines	By the end of the session, the trainee should be able to: 1. Identify key workplace policies and guidelines relevant to electrical installation work. 2. Explain the	- Safety policies, quality control, environmental policies - Reporting procedures and chain of command - Professional	- Engage in a Case Study of a simulated ethical dilemma in a workplace setting, requiring trainees to refer to a given code of conduct. - Participate in a	- Sample workplace policies and code of conduct - Ethical dilemma	Knowledge Checks: -Oral questioning on policy adherence. -Oral	

			importance of observing organizational policies and codes of conduct. 3. Apply ethical decision-making principles in situations involving workplace policies and codes.	behavior, integrity, confidentiality - Avoiding conflicts of interest - Dilemmas in compliance - Impact of non-compliance	Group Discussion on the consequences of ignoring workplace policies, both for the individual and the organization. - Conduct a Q&A Sessions to clarify specific policies or ethical standards relevant to their future roles. <u>Follow up Activity:</u> 1. Review a sample workplace code of conduct and highlight three clauses most relevant to their daily work as an electrician. (5 Marks) Due date:	scenarios	questioning -Written response to an ethical scenario. Attitudes: -Integrity in professional practice. -Respect for organizational rules.	
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10	1&2	Ethical Work Practices - Professionalism and Values	By the end of the session, the trainee should be able to: 1. Define professionalism and its manifestations in the electrical trade. 2. Explain how personal goals align with organizational policies to foster professionalism. 3. Demonstrate professional conduct and uphold self-worth in various workplace	- Appearance, punctuality, communication skills - Dedication to quality work - How personal aspirations can be met through professional growth - Contributing to organizational success - Valuing one's skills and contributions - Advocating for fair treatment and recognition	- Engage in a Think-Pair-Share activity to brainstorm characteristics of a 'professional electrician' and the impact of professionalism. - Participate in a Role Play simulating a client interaction, focusing on professional communication and problem-solving. - Conduct a Group Discussion on how maintaining self-worth influences one's performance and	- Articles on professionalism in trades - Scenarios for professional interaction	Knowledge Checks: -Oral discussion on professional ethics. -Written self-assessment on professional conduct. -Oral Discussion Attitudes: -Commitment to excellence.	
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			interactions.		perception in the workplace. <u>Follow up Activity:</u> 1. Outline three personal professional goals for the next year and explain how they align with a professional code of conduct. (10 Marks) Due date:		-Pride in one's work.	
10	3&4	Ethical Work Practices - Teamwork and Conflict Resolution	By the end of the session, the trainee should be able to: 1. Explain the principles of effective teamwork in collaborative electrical installation projects.	- Communication, cooperation, shared responsibility - Synergy in project execution	- Engage in a Jigsaw activity where groups learn different aspects of effective teamwork and teach them to others.	- Team-building exercises - Conflict resolution frameworks - Sample organizational policies on	Knowledge Checks: -Oral questioning on teamwork principles.	

			2. Identify common sources of conflict within a team and their potential impact. 3. Apply constructive conflict resolution strategies in line with organizational policies.	- Differing opinions, work styles, resource allocation - Miscommunication - Active listening, mediation, negotiation - Escalation procedures (if necessary)	- Participate in a Role Play simulating a team conflict scenario, practicing negotiation and resolution skills. - Conduct a Group Discussion on how strong teamwork directly impacts safety and efficiency in complex electrical installations. <u>Follow up Activity:</u> 1. Describe a hypothetical team conflict in a workplace and propose a solution adhering to workplace policies. (10	grievance handling	-Written response to conflict scenario. -Oral Discussion Attitudes: 1. Collaborative spirit. 2. Constructive approach to disagreement.	
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					Marks) Due date:			
	5&6	Review of Identifying Trunking Components	By the end of the session, the trainee should be able to: 1. Accurately identify all major trunking types, sizes, and accessories. 2. Differentiate components based on their material, application, and IEC standards. 3. Select appropriate components for varied installation	- PVC, Metallic, Galvanized Iron - Standard dimensions (16x16, 25x16, etc.) - Sockets, switches, lamp holders, joint boxes - Pattress, circuit breakers, CCU, cut-outs, energy meters, earth rods, switch fuse - Matching components to specific project needs - Ensuring	- Engage in a Quiz Games session to rapidly identify trunking types, sizes, and accessories from visual cues. - Participate in a Gallery Walk showcasing a variety of trunking components, with trainees labeling and describing their uses. - Conduct a Case Study analysis of an installation plan, where trainees must	- Wide range of trunking samples and accessories - IEC standards documentation - Installation project plans	Knowledge Checks: -Oral identification of components. -Written component selection exercise. -Oral questioning Attitudes: -Thoroughness	

			environments and requirements.	compliance with safety and performance standards	justify their selection of trunking types, sizes, and accessories. <u>Follow up Activity:</u> 1. Compile a detailed inventory list for a small office trunking installation, specifying component types, sizes, and quantities. (15 Marks) Due date:		in component recognition. -Commitment to correct material specification.	
10	7&8	Review of Preparing Trunking Work Pieces	By the end of the session, the trainee should be able to: 1. Reiterate essential safety measures for all preparation tasks.	- OSHA and EHS compliance reminders - Correct PPE usage and hazard awareness	- Engage in a Round Robin where each trainee presents a key safety tip for trunking preparation and	- Trunking materials (PVC, metallic, GI) - Full range of preparation	Knowledge Checks: -Oral questioning on safety and tool usage.	

			2. Demonstrate proficiency in measuring, marking, and cutting various trunking materials. 3. Reveal proper housekeeping and tool maintenance practices.	- Accurate measurement and marking techniques - Clean and precise cutting of PVC, metallic, and GI trunking - Efficient cleanup procedures - Proper tool storage and care	explains its importance. - Participate in a Demonstrations with Participation on cutting and deburring different trunking materials, focusing on speed and accuracy. - Conduct a Group Discussion on best practices for maintaining a safe and organized workstation during preparation tasks. <u>Follow up Activity:</u> 1. On a practical station, prepare a mock trunking	tools - Safety guidelines	-Practical demonstration of preparation skills. -Practical demonstration -Oral questioning Attitudes: -Responsibility for safety. -Precision in craftsmanship.	
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					corner (measure, mark, cut, deburr) demonstrating all learned techniques. (20 Marks) Due date:			
11	1&2	Comprehensive Review of Mounting Trunking Work Pieces	By the end of the session, the trainee should be able to: 1. Accurately mark mounting points according to IEC standards and installation plans. 2. Demonstrate proficiency in mounting PVC, metallic, and galvanized iron	- Interpreting plans, applying IEC spacing rules - Using levels and marking tools effectively - Fastening methods for different trunking types and surfaces	- Engage in a Case Study where trainees analyze a complex mounting scenario and propose a detailed installation plan, including marking and support. - Participate in a Demonstrations with Participation session focused	- Mock-up walls/surfaces - All trunking types and mounting hardware - Measuring and marking tools - IEC standards	Knowledge Checks: -Oral questioning -Practical assessment of mounting techniques.	

			trunking.	- Ensuring levelness and secure attachment	on demonstrating the correct technique for mounting trunking at a 90-degree internal corner. - Conduct a Think-Pair-Share activity to discuss common mounting errors and their preventative measures. <u>Follow up Activity:</u> 1. On a mock-up wall, mount a section of trunking that includes both a straight run and a 45-degree bend, ensuring adherence to standards. (25 Marks) Due date:		-Oral questioning on support systems Attitudes: 1. Dedication to structural integrity. 2. Aesthetic sensibility in installation.	
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11	3&4	Review of Installing Electrical Cables and Accessories (Part 1)	By the end of the session, the trainee should be able to: 1. Accurately identify cable colour codes and select appropriate cable sizes for various circuits. 2. Demonstrate safe and neat cable laying techniques within trunking. 3. Review the installation of single-phase intake point components (cut-outs, meters, CCUs).	- IEC colour codes for phases, neutral, earth - Matching cable size to load and protection - Avoiding damage, maintaining bending radii - Dressing and routing for neatness - Function and connection of cut-outs, energy meters, CCUs - Earthing and switch fuse installation	- Engage in a Quiz Games session to test knowledge on cable colour codes, sizes, and their applications. - Participate in a Demonstrations with Participation on efficient cable laying and dressing techniques in a trunking run. - Conduct a Group Discussion on the sequential steps and safety considerations for installing a complete single-	- Various cable samples - Mounted trunking sections - Cut-outs, energy meters, CCUs, switch fuses, earth rod (mock) - Wiring diagrams	Knowledge Checks: -Oral questioning on cable sizing. -Practical demonstration of cable laying techniques -Practical demonstration of cable installation and intake point wiring. Attitudes:	
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					phase intake point. <u>Follow up Activity:</u> 1. On a mock-up, install a 1-phase cut-out and connect it to a CCU, using correctly sized and colored cables within trunking. (30 Marks) Due date:		-Diligence in adhering to wiring regulations. -Proactive safety awareness.	
	5&6	Review of Installing Electrical Cables and Accessories (Part 2)	By the end of the session, the trainee should be able to: 1. Revisit the installation of various lighting circuits (one-way, two-way, intermediate).	- Wiring diagrams and control logic for all types - Ensuring functional operation	- Engage in a Peer Teaching session where trainees demonstrate and explain the wiring of a two-way lighting circuit.	- Mock-up boards with trunking for all circuit types - Switches, sockets, lamp holders,	Knowledge Checks: -Practical assessment of lighting circuit installation.	

			2. Review looping-in methods at ceiling roses, joint boxes, and switches. 3. Practice the installation of power circuits including socket outlets, cooker units, and water heaters.	- Advantages and implementation at different points - Ensuring secure and accessible connections - Ring vs. radial circuits, appropriate protective devices - High-power appliance connections (cooker, water heater)	- Participate in a Round Robin activity, rotating through stations to practice different looping-in connections and power circuit wiring. - Conduct a Case Study of complex wiring scenarios, requiring trainees to identify the appropriate circuit type and installation method. <u>Follow up Activity:</u> 1. Install and test a functional intermediate lighting circuit on a mock-up board, ensuring all looping-in	cooker units, water heater isolators - Electrical cables - Comprehensive wiring diagrams	-Oral questioning on complex wiring logic. -Practical assessment of power circuit installation. Attitudes: 1. Mastery of diverse wiring techniques. 2. Creative problem-solving for circuit design.	
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					connections are correct. (35 Marks) Due date:			
11	7&8	Review on Performing Tests and Inspection	By the end of the session, the trainee should be able to: 1. Revisit systematic visual inspection procedures for electrical installations. 2. Demonstrate proficiency in performing continuity tests on protective conductors and circuits.	- Key checkpoints for compliance and safety - Documenting observations - Earth continuity, equipotential bonding - Ring final circuit continuity	- Engage in a Gallery Walk of mock installations (with intentional faults), where trainees perform visual inspections and identify issues. - Participate in a Demonstrations with Participation session focused on safely and accurately performing all major electrical tests (continuity, polarity, ring) on a complete mock	- Fully wired mock-up installation with deliberate faults - Multi-function electrical tester - Test certificates/forms - IEC standards for testing	Knowledge Checks: -Practical assessment on visual inspection procedures. -Practical assessment on continuity tests procedures.	

			3. Practice polarity and ring test procedures accurately and interpret results.	- Importance of correct polarity - Detailed ring circuit test procedures (R1+R2, R1, Rn) - Interpreting all test results against standards	circuit. - Conduct a Group Discussion on the legal and safety implications of incomplete or incorrect testing procedures. <u>Follow up Activity:</u> 1. Perform a complete set of initial verification tests (visual, continuity, polarity, ring) on a mock-up installation and fill out a simulated test certificate. (40 Marks) Due date:		-Written interpretation of test results and fault diagnosis. Attitudes: -Rigorous adherence to testing protocols. -Unwavering commitment to safety verification.	
12	1&2	Final Preparatory	By the end of the session, the				Knowledge Checks:	

		Session 1 - Overall Unit Review and Q&A	trainee should be able to: 1. Consolidate knowledge on identifying trunking components and preparing work pieces. 2. Reinforce understanding of various trunking mounting techniques and support systems. 3. Address any remaining queries or areas of confusion across all topics covered.	- Comprehensive recap of LO1 and LO2 content. - Important standards and safety practices. - Common challenges in preparation and mounting. - Efficient workflow strategies. - Open forum for trainee questions. - Targeted revision based on identified knowledge gaps.	- Engage in a Crowdsourcing session where trainees submit their most challenging questions on LO1, LO2, and LO3 for group discussion. - Participate in a Quiz Games focusing on rapid recall of facts and procedures related to trunking identification, preparation, and mounting. - Conduct a Group Discussions to elaborate on complex topics and clarify misconceptions.	- Full curriculum outline - Previous quizzes/tests - Whiteboard/fli pchart for questions	-Oral questioning -Short diagnostic quiz. -Oral questioning Attitudes: -Proactive engagement in revision. -Confidence in	
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					<u>Follow up Activity:</u> 1. Complete a self-assessment questionnaire covering LO1, LO2, and LO3, identifying areas for final review before the CAT. (Not Graded) Due date:		acquired knowledge.	
12	3&4	Final Preparatory Session 2 - Installation and Testing Review	By the end of the session, the trainee should be able to: 1. Consolidate knowledge on installing electrical cables, accessories, and complete circuits. 2. Reinforce understanding of all testing	- Cable sizing, laying, intake points, lighting, and power circuits. - Looping-in methods and high-power connections. - Visual inspection, continuity,	- Engage in a Case Study analysis of a simulated faulty electrical installation, requiring trainees to diagnose and propose solutions. - Participate in a Round Robin activity where	- Mock-up installation with various circuits - Test equipment - Fault diagnosis scenarios	Knowledge Checks: -Oral questioning on fault-finding. -Practical demonstration of key	

			and inspection procedures.	polarity, ring tests. - Interpreting test results and fault diagnosis.	trainees perform quick practical demonstrations of key installation and testing steps.		installation steps.	
			3. Practice critical thinking and problem-solving skills for complex installation and fault diagnosis scenarios.	- Applying knowledge to hypothetical installation failures. - Developing troubleshooting methodologies.	- Conduct a Brainstorming session on potential issues that could arise during a practical assessment and how to systematically address them. <u>Follow up Activity:</u> 1. Prepare a list of key safety checks to perform before, during, and after any electrical installation project. (10 Marks) Due date:		-Practical demonstration Attitudes: -Strategic approach to problem-solving. -Confidence in practical skills.	

	5&6	Final Preparatory Session 3 - Ethical Practices and Practical Readiness	By the end of the session, the trainee should be able to: 1. Review ethical work practices, professionalism, and teamwork principles. 2. Ensure practical stations are set up and tools are ready for the final assessment. 3. Conduct a final briefing on CAT 3 procedures and expectations.	- Personal management, policies, code of conduct, teamwork, conflict resolution. - Importance of professional behavior during assessment. - Checking availability and condition of tools, materials, and PPE. - Familiarity with the assessment environment. - CAT 3 - Reiteration of assessment scope, rules, and marking criteria. - Time	- Engage in a Role Play where trainees demonstrate effective teamwork and conflict resolution in a practical task setting. - Participate in a Demonstrations with Participation session on the pre-assessment check of tools and PPE. - Conduct a Q&A Sessions to clarify any administrative or procedural	- Relevant ethical guidelines - Tools and PPE - CAT 3 instructions and rubrics	Knowledge Checks: -Oral discussion on ethical dilemmas. -Practical check of tool readiness. -Oral discussion Attitudes: - Professionalis	
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				management and safety instructions.	aspects of the upcoming final CAT. <u>Follow up Activity:</u> 1. Ensure all personal tools are cleaned, checked, and ready for the final assessment. (Not Graded) Due date:		m in all interactions. -Diligence in preparation.	
12	7&8	Continuous Assessment Test (CAT) 3 - Final Practical and Theory	By the end of the session, the trainee should be able to: 1. Demonstrate comprehensive practical competence in installing and testing a complete trunking system.	- Comprehensive evaluation of all learning outcomes (LO1-LO5 and Ethical Practices). - Integrated practical installation and testing tasks.	- Independently plan and execute a comprehensive trunking installation and testing project according to given specifications.	- Designated practical assessment bay with all required materials - Comprehensive written examination	Knowledge Checks: 1. Holistic practical assessment of unit competence. 2. Comprehensive written and oral examination. Attitudes: 1. Utmost professionalism and	

			2. Apply theoretical knowledge to justify installation decisions and troubleshoot simulated faults effectively.	- Major practical project (e.g., install a complete mini-trunking circuit system including intake, lighting, power, and testing). - Written component covering theoretical concepts and critical decision-making.	- Complete a written examination demonstrating theoretical understanding and problem-solving abilities.	paper - All necessary tools and PPE - Assessment rubrics and safety guidelines	integrity. 2. Self-reliance and meticulousness .	
			3. Adhere to all safety regulations, industry standards, and ethical practices during a final, holistic assessment.	- Marking based on adherence to standards, safety, functionality, neatness, and efficiency. - Oral questioning during practical assessment.	- Maintain strict adherence to all safety protocols and assessment instructions throughout the duration of the test. <u>Follow up Activity:</u> 1. Await assessment			

					feedback and reflect on overall learning journey and future professional development. (Not Graded) Due date:			
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